

Q A 4×4 , 3 bits / pixel image is given by

(21)

$$f(m,n) = \begin{bmatrix} 0 & 2 & 4 & 5 \\ 7 & 3 & 6 & 2 \\ 3 & 2 & 1 & 0 \\ 5 & 6 & 1 & 7 \end{bmatrix}$$

Sketch its ~~two~~ bit planes.

Soln:- The three bit representation of the given values is

0 - 000

4 - 100

1 - 001

5 - 101

2 - 010

6 - 110

3 - 011

7 - 111

Bit Plane 0 - obtained by taking LSB values of the given no.

$$\text{Bit Plane '0'} = \begin{bmatrix} 0 & 0 & 0 & 1 \\ 1 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 \\ 1 & 0 & 1 & 1 \end{bmatrix}$$

Bit Plane 1 - obtained by taking next LSB values of given no.

$$\text{Bit Plane '1'} = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 1 & 1 & 1 & 1 \\ 1 & 1 & 0 & 0 \\ 0 & 1 & 0 & 1 \end{bmatrix}$$

Bit Plane 2 - obtained by taking MSB value of given no.

$$\text{Bit Plane '2'} = \begin{bmatrix} 0 & 0 & 1 & 1 \\ 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 \end{bmatrix}$$



Histogram

- Histogram techniques are very effective and useful in many image processing applications
- It is an effective tool not only for image quality assessment, but also for manipulating the contrast and brightness of an image
- The histogram of an image is a plot of the number of occurrences of gray level in the image against the gray-level values.
- It can be visualized as an intensity distribution or a probability density function.
- It provides more insight about image contrast & brightness
 1. The histogram of dark image will be clustered towards the lower gray level.
 2. The histogram of bright image will be clustered towards the higher gray level.
 3. For low-contrast image, the histogram will not spread equally, the histogram will be narrow
 4. For higher contrast image, the histogram will have an equal spread in the gray level.
- Histogram is a discrete function formed by counting the number of pixels that have certain gray level in the image.
- Histogram of a digital image with gray levels in the range $[0, L-1]$ is a discrete function

$$h(r_k) = n_k$$

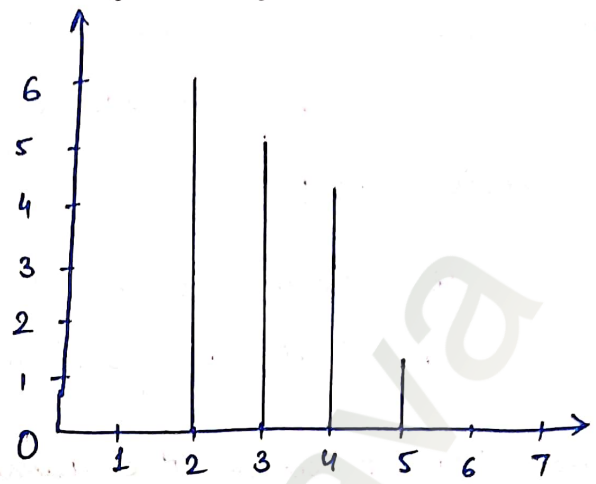
where $r_k \rightarrow k^{\text{th}}$ gray level

$n_k \rightarrow$ number of pixels in the image having gray level r_k

$h(r_k) \rightarrow$ Histogram of a digital image with gray levels r_k

Q Draw the histogram of the following image.

2	3	3	2
4	2	4	3
3	2	3	5
2	4	2	4



Normalized Histogram -

- Dividing each of the histogram at gray level (r_k) by the total number of pixels in the image (n).

$$p(r_k) = \frac{n_k}{n}$$

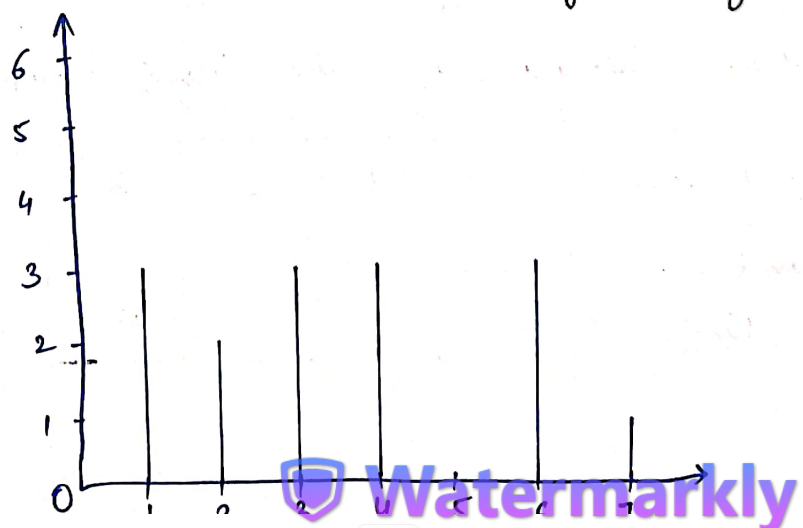
for $k = 0 \dots L-1$

$p(r_k)$ = estimate of the probability of gray level occurrence of r_k .

Note:- Sum of all components of a normalized histogram is equal to 1.

Q An image with gray level b/w 0 to 7. Find histogram of the image & normalize it.

1	6	2	2
1	3	3	3
4	6	4	0
1	6	4	7



Histogram Equalisation

- Equalisation is a process that attempts to spread out the gray levels in an image so that they are evenly distributed across their range.
- It is a technique where the histogram of the resultant image is as flat as possible.
- Procedure to perform Histograms Equalization

Step 1. Find the running sum of the histogram values.

Step 2. Normalize the values from Step 1 by dividing the number of pixels.

Step 3. Multiply the values from Step 2 by maximum gray level value

Step 4. Map the gray values to the results from Step 3 using a one to one correspondance.

Q Perform histogram equalisation of the image.

$$\begin{bmatrix} 4 & 4 & 4 & 4 & 4 \\ 3 & 4 & 5 & 4 & 3 \\ 3 & 5 & 5 & 5 & 3 \\ 3 & 4 & 5 & 4 & 3 \\ 4 & 4 & 4 & 4 & 4 \end{bmatrix}$$

Soln:-

Step 1. Histogram of the input image is given as -

Gray levels	0	1	2	3	4	5	6	7
Number of pixels	0	0	0	6	14	5	0	0

Step 1. Compute the running sum of histogram values.

Note: Running sum is k/a Cumulative distribution function.

Gray level	0	1	2	3	4	5	6	7
No. of Pixel	0	0	0	6	14	5	0	0
Running Sum	0	0	0	6	20	25	25	25

Step 2. Divide the running sum by total number of pixels

Gray level	0	1	2	3	4	5	6	7
No. of Pixel	0	0	0	6	14	5	0	0
Running Sum	0	0	0	6	20	25	25	25
RS / Total No. of Pixel	$\frac{0}{25}$	$\frac{0}{25}$	$\frac{0}{25}$	$\frac{6}{25}$	$\frac{20}{25}$	$\frac{25}{25}$	$\frac{25}{25}$	$\frac{25}{25}$

Step 3. Multiply the values from max. gray level.

Gray level	0	1	2	3	4	5	6	7
No. of Pixel	0	0	0	6	14	5	0	0
Running Sum	0	0	0	6	20	25	25	25
RS / Total No. of Pixel	$\frac{0}{25}$	$\frac{0}{25}$	$\frac{0}{25}$	$\frac{6}{25}$	$\frac{20}{25}$	$\frac{25}{25}$	$\frac{25}{25}$	$\frac{25}{25}$
Multiply	$\frac{0}{25} \times 7$	$\frac{0}{25} \times 7$	$\frac{0}{25} \times 7$	$\frac{6}{25} \times 7$	$\frac{20}{25} \times 7$	$\frac{25}{25} \times 7$	$\frac{25}{25} \times 7$	$\frac{25}{25} \times 7$
Round off	0	0	0	2	6	7	7	7

Step 4. Mapping of gray level by one-to-one

Original Gray level	Histogram Equalised Values
0	0
1	0
2	0
3	2
4	6
5	7
6	7
7	7

Therefore, the original image and histogram equalised image is as shown

$$\begin{bmatrix} 4 & 4 & 4 & 4 & 4 \\ 3 & 4 & 5 & 4 & 3 \\ 3 & 5 & 5 & 5 & 3 \\ 3 & 4 & 5 & 4 & 3 \\ 4 & 4 & 4 & 4 & 4 \end{bmatrix} \xrightarrow[\text{Equalisation}]{\text{Histogram}} \begin{bmatrix} 6 & 6 & 6 & 6 & 6 \\ 2 & 6 & 7 & 6 & 2 \\ 2 & 7 & 7 & 7 & 2 \\ 2 & 6 & 7 & 6 & 2 \\ 6 & 6 & 6 & 6 & 6 \end{bmatrix}$$

Original Image

Histogram Equalised Image

Q Perform Histogram Equalisation of the image

2	3	3	2
4	2	4	3
3	2	3	5
2	4	2	4

4x4 Image

Gray Scale = [0, 9]

Soln:- Histogram of the input image is

(24)

Gray level	0	1	2	3	4	5	6	7	8	9
No. of Pixel	0	0	6	5	4	1	0	0	0	0

Step 1. Compute the running sum

Gray level	0	1	2	3	4	5	6	7	8	9
No. of Pixel	0	0	6	5	4	1	0	0	0	0
Running Sum	0	0	6	11	15	16	16	16	16	16

Step 2. Divide the running sum by total no. of pixels

Gray level	0	1	2	3	4	5	6	7	8	9
No. of Pixel	0	0	6	5	4	1	0	0	0	0
Running Sum	0	0	6	11	15	16	16	16	16	16
RS / No. of p.	$\frac{0}{16}$	$\frac{0}{16}$	$\frac{6}{16}$	$\frac{11}{16}$	$\frac{15}{16}$	$\frac{16}{16}$	$\frac{16}{16}$	$\frac{16}{16}$	$\frac{16}{16}$	$\frac{16}{16}$

Step 3. Multiply the values from max. gray level

Gray level	0	1	2	3	4	5	6	7	8	9
No. of Pixel	0	0	6	5	4	1	0	0	0	0
RS	0	0	6	11	15	16	16	16	16	16
RS / TNOP	$\frac{0}{16}$	$\frac{0}{16}$	$\frac{6}{16}$	$\frac{11}{16}$	$\frac{15}{16}$	$\frac{16}{16}$	$\frac{16}{16}$	$\frac{16}{16}$	$\frac{16}{16}$	$\frac{16}{16}$

Multiply

$$\frac{0}{16} \times 9 \quad \frac{0}{16} \times 9$$

Round off

	0	0	3	6	8	9	9	9	9	9
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Step 4. Mapping of gray level

Original Gray level

0
1
2
3
4
5
6
7
8
9

Histogram Equalised

0
0
3
6
8
9
9
9
9
9

Therefore,

2	3	3	2
4	2	4	3
3	2	3	5
2	4	2	4

Histogram
Equalisation

3	6	6	3
8	3	8	6
6	3	6	9
3	8	3	8

Original Image

eg

1	3	5
4	4	3
5	2	2

Histogram Equalised Image

1	4	7
5	5	4
7	2	2

eg

1	2	3	4
5	5	6	6
6	7	6	6
6	7	2	3

0	1	2	3
4	4	6	6
6	7	6	6
6	7	1	2

Q. Suppose a 3-bit image of size 64×64 pixels ($MN = 4096$) has the intensity distribution shown in the following table. Get the histogram equalization transformation function and give the $p_s(s_k)$ for each s_k . (25)

r_k	n_k	$p_r(r_k) = n_k / MN$
$r_0 = 0$	790	0.19
$r_1 = 1$	1023	0.25
2	850	0.21
3	656	0.16
4	329	0.08
5	245	0.06
6	122	0.03
7	81	0.02

Note:- Normalisation $p_r(r_k) = \frac{n_k}{MN}$

Discrete form of transformation,

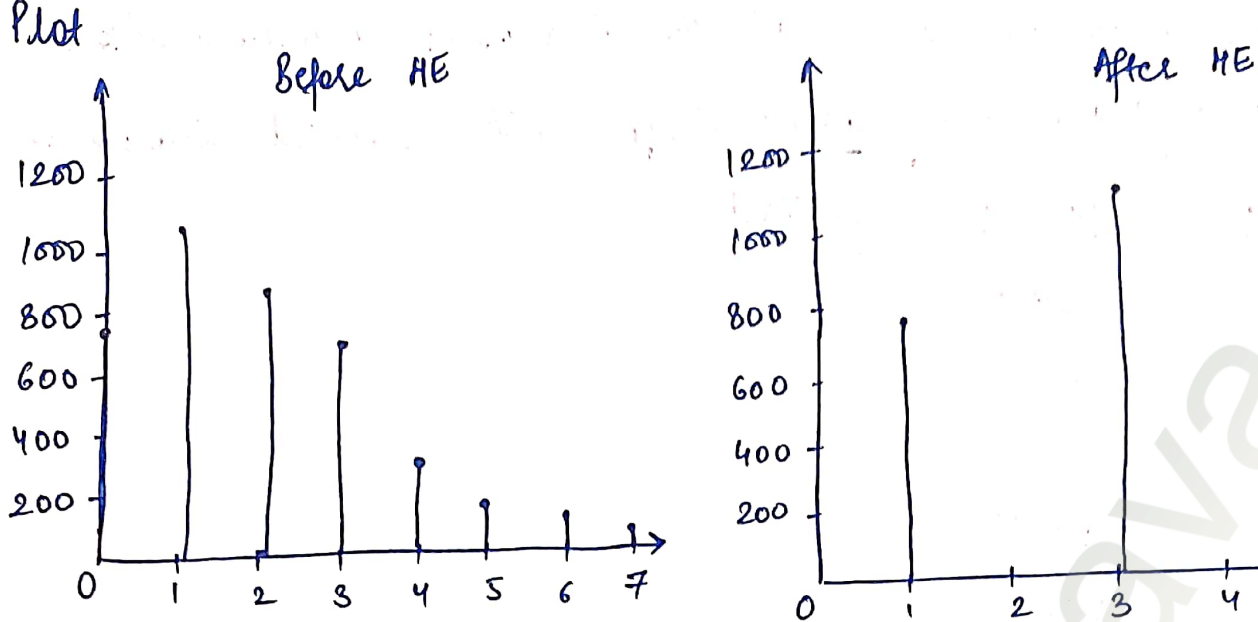
$$s_k = T(r_k) = (L-1) \sum_{j=0}^k p_r(r_j), \quad k=0, 1, \dots, L-1$$

Soln:

x_k	n_k	$P_k(x_k)$	Running Sum S_k	x_k^* S_k	Round off
0	790	0.19	0.19	1.33	1
1	1023	0.25	0.44	3.08	3
2	850	0.21	0.65	4.55	5
3	656	0.16	0.81	5.67	6
4	329	0.08	0.89	6.23	6
5	245	0.06	0.95	6.65	7
6	122	0.03	0.98	6.86	7
7	81	0.02	1	7	7

Equalised

0	0
1	790
2	0
3	1023
4	0
5	850
6	$656 + 329 = 985$
7	$245 + 122 + 81 = 448$



Histogram Matching (Specification)

- Histogram equalisation produces a transformation function that generates an output image with uniform histogram.
- When automatic enhancement is desired, it is a good approach because the results are predictable and method is simple to implement.
- Sometimes it is useful to specify the shape of the histogram that we wish to the processed image have.
- The method used to generate images that have specified histogram is called histogram matching or histogram specification.

Process -

- Step 1. Find histogram of input image and find equalization
 - Step 2. Specify the desired histogram, and find histogram equalisation
 - Step 3. Build look up table
- for each gray level k , find S_k and then find a level such that S_k best matches Z_q

Min $|S_k - Z_q|$

Q Repeat the above example, get the histogram transformation function and make the output image with specified histogram.

Z_q	$p_q(Z_q)$	Specified
0	0.00	
1	0.00	
2	0.00	
3	0.15	
4	0.20	
5	0.30	
6	0.20	
7	0.15	

Soln:-

x_k	n_k	$P(n_k)$	Σ	$(L-1)\Sigma$	S_k
0	790	0.19	0.19	1.33	1
1	1023	0.25	0.44	3.08	3
2	850	0.21	0.65	4.55	5
3	656	0.16	0.81	5.67	6
4	329	0.08	0.89	6.23	6
5	245	0.06	0.95	6.65	7
6	122	0.03	0.98	6.86	7
7	81	0.02	1	1	7

Z_k	$P(Z_k)$	Σ	$(L-1)\Sigma$	$G(Z_k)$
0	0.00	0	0	0
1	0.00	0	0	0
2	0.00	0	0	0
3	0.15	0.15	0.05	1
4	0.20	0.35	2.45	2
5	0.30	0.65	4.55	5
6	0.20	0.85	5.95	6
7	0.15	1	7	7

k	s_k	$G(z_k)$	z_k
0	1	0	0
1	3	0	1
2	5	0	2
3	6	1	3
4	6	2	4
5	7	5	5
6	7	6	6
7	7	7	7

• Mapping of values s_k into corresponding z_q

s_k	$\longrightarrow$	z_q
1	$\longrightarrow$	3
3	$\longrightarrow$	4
5	$\longrightarrow$	5
6	$\longrightarrow$	6
7	$\longrightarrow$	7

eg Perform histogram specification on 8×8 , eight level image described as

P_k	8	10	10	2	12	16	4	2
s_k	0	1	2	3	4	5	6	7

The -target histogram is

s_k	0	1	2	3	4	5	6	7
S_k	0	0	0	0	20	20	16	8

Soln:-

Equalization Process

r_k	P_k	Σ	$\frac{\Sigma}{\text{Total}} \times L-1$	Roundoff
0	8	8	$\frac{8}{64} \times 7 = 0.875$	1
1	10	18	$\frac{18}{64} \times 7 = 1.968$	2
2	10	28	$\frac{28}{64} \times 7 = 3.0625$	3
3	2	30	$\frac{30}{64} \times 7 = 3.2812$	3
4	12	42	$\frac{42}{64} \times 7 = 4.5937$	5
5	16	58	$\frac{58}{64} \times 7 = 6.3437$	6
6	4	62	$\frac{62}{64} \times 7 = 6.7125$	7
7	2	64	$\frac{64}{64} \times 7 = 7$	7

Target

k_k	S_k	Σ	$\Sigma / \text{Total} \times L-1$	Round off
0	0	0	0	0
1	0	0	0	0
2	0	0	0	0
3	0	0	0	0
4	20	20	$20/64 \times 7 = 2.1875$	2
5	20	40	$40/64 \times 7 = 4.375$	4
6	16	56	$56/64 \times 7 = 6.125$	6
7	8	64	$64/64 \times 7 = 7$	7

k_k	S_k	$G(Z_k)$	Z_k
0	1	0	0
1	2	0	1
2	3	0	2
3	3	0	3
4	5	2	4
5	6	4	5
6	7	6	6
7	7	7	7

